Reading structured data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

Teach each aggregate together with its data policy.

Supply complete named-field fixtures, preserve required-field errors and distinguish offered rows from measured rows.

Keep one habit: Ask which rows contributed, why zero was included and what the missing values still leave unknown.

MAIN LESSON · SECTION 1

Prepare the connected bridge lesson

> Check prior exposure and keep execution practice distinct from paper reasoning.

Goal and main plan

Primary target: **Process** a list of records consistently

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Retrieval and prerequisite guidance

Without opening the old lesson, explain one key idea from Dictionaries and records. Give an example and a mistake that the idea helps you avoid.

Look for: Read and update named dictionary **field**s

Reference example, not the only acceptable wording: Keys name, capacity, used; used is 2; remaining spaces 7.

Functions and arguments

Without opening the old lesson, explain one key idea from Functions and arguments. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a function definition from a function call

Reference example, not the only acceptable wording: Name show_remaining; parameters capacity and used. Arguments 15 and 6 match capacity = 15 and used = 6. Expected printed result 9.

Branching and boundary tests

Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

Look for: Trace the selected branch in an if elif else chain

Reference example, not the only acceptable wording: if tests negative first; elif checks <= 12 only after that fails; else handles the rest.

Unindented print runs after any selected branch, whose assignment gives label a value.

Teach Week 11 as a two-session investigation using fictional data. Two fictional cupboards are stored as a list of dictionaries. Each record uses the same keys and units. We want the total spare spaces.

Treat the records as a **dataset**, and trace one record at a time before trusting a total. Assess the three stated outcomes through the independent task, not by copying the worked result.

Materials and readiness

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

The key "used" stores 3.

End goal

Trace reading structured data and explain the changed program before running it.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Dictionaries and records

Outcomes

Process a list of records consistently

Compute a total from named **field**s

Preserve record meaning while tracing a loop

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach aggregation with an explicit row policy

> Required cupboard fields and optional observed counts are different contracts.

Read required-field cupboard data and empty-input limit

Two fictional cupboards are stored as a list of dictionaries. Each record uses the same keys and units. We want the total spare spaces. Treat the records as a **dataset**, and trace one record at a time before trusting a total.

The loop variable row refers to each record in turn. We calculate spare from that record, then add it. total starts before the loop.

An empty list produces 0 because there are no updates; it does not tell us whether no cupboards exist or records were not collected.

Supply complete W2 record inputs

```
rows = [{"capacity": 4, "used": 2},
        {"capacity": 7, "used": 3},
        {"capacity": 2, "used": 1}]
total = 0
for row in rows:
    spare = row["capacity"] - row["used"]
    total = total + spare
print(total)
```

Read the pictured inclusion policy

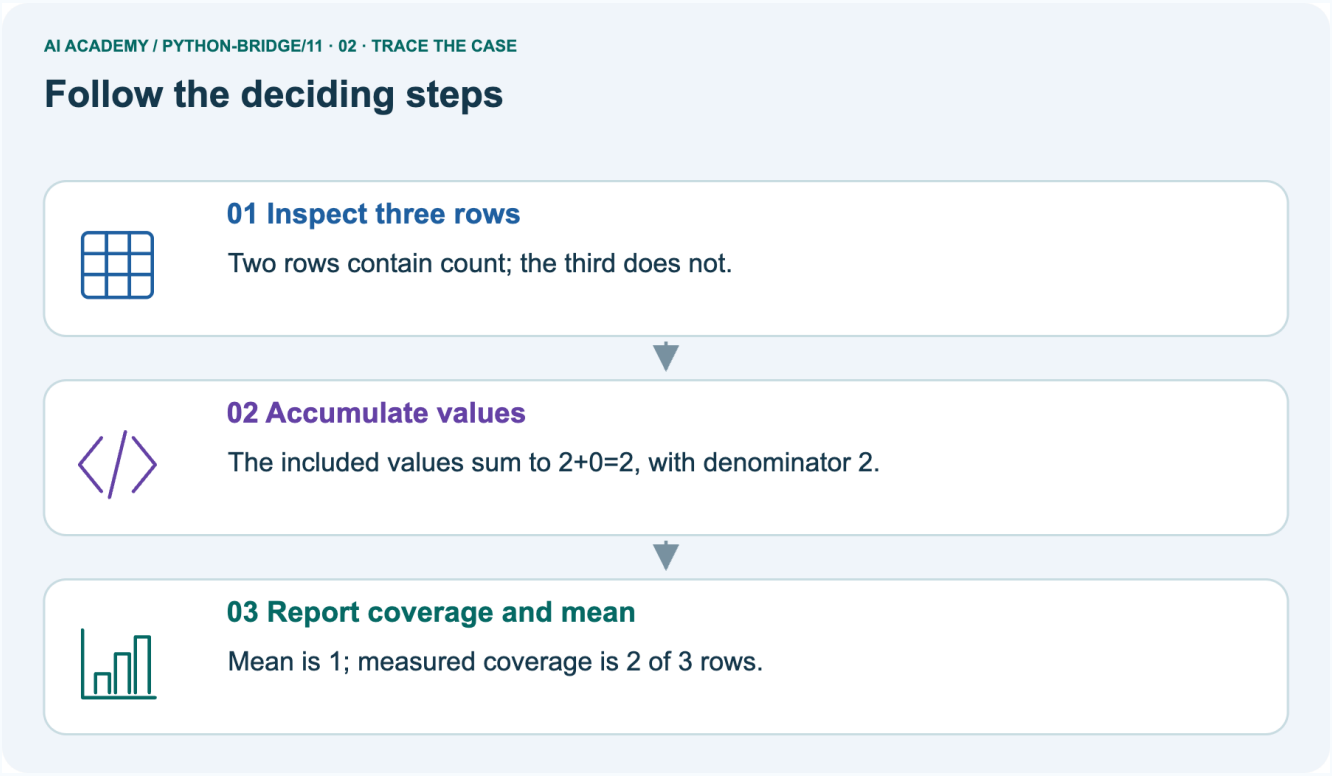
Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.

Python 3 records: [{"count":2},{"count":0},{}]. Policy: include existing numeric counts, including zero; skip an absent **field**.

The supplied rows contain count 2, count 0 and an absent field. append adds one value to the end of observed, retaining the real zero. sum(observed)adds included values; len(observed) is the mean denominator, while len(rows) is the number of offered records. With values 2 and 0, sum 2/denominator 2 gives mean 1.0 and coverage 2/3. None or an absent field is unavailable under this toy policy.

The program assumes supplied numeric values; it does not reject malformed types or verify measurements. An observed empty sum 0 does not prove missing population values are zero.

Trace zero, missing and coverage



Follow the labelled picture with the matching explanation. Open the vector diagram

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Included values and absent field

	Count value	Include in mean?
Row 1	2	Yes
Row 2	0	Yes
Row 3	Absent	No

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Inspect three rows	Two rows contain count; the third does not.
2. Accumulate values	The included values sum to 2+0=2, with denominator 2.
3. Report coverage and mean	Mean is 1; measured coverage is 2 of 3 rows.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Structured data need a row policy before aggregation. Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an **observation**.

```
rows = [{"count": 2}, {"count": 0}, {}]
observed = []
for row in rows:
    value = row.get("count")
    if value is not None:
        observed.append(value)
observed_total = sum(observed)
if len(observed) > 0:
    mean = observed_total / len(observed)
else:
    mean = None
print(observed_total, len(observed), len(rows), mean)
```

Predict adding an observed fourth row

Change: Add {"count":4} as a fourth row.

Prediction: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

Reason: Only the added recorded value contributes; the existing absent **field** remains skipped.

State parsing and provenance limits

This worked row policy is explicit; file parsing, malformed values and data provenance require additional checks.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Additional guided key

Contributions 5, 0, 6; total 11. Exit: No. No records were **process**ed; the result provides no **evidence** about actual cupboards.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Dataset	A collection of records
Row	One record in a table
Field	One named part of a record
Accumulator	A variable updated to keep a running result

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
First row	8 - 3 = 5	total becomes 5
Second row	6 - 6 = 0	total stays 5

Read each row from left to right and connect the evidence to the stated rule.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **Add a third record with capacity 10 and used 4. Predict the total and list each contribution.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Reading structured data

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the trace, then fade the help

> Collect predictions before execution or feedback.

Main completion task and key

Completion task: For spare contributions 2, 4 and 1, trace total starting at 0.

Teacher check: Running totals 2, 6, 7.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Reading structured data. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: This loop assumes every record has the required integer **field**s. A missing used **field** causes KeyError. Do not silently treat a missing count as zero; that would invent unused capacity.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case key

Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

Initially values 2 and 0 give sum 2 and mean 1 across two **observation**s, coverage 2/3. Adding 4 gives sum 6, mean 2 across three **observation**s, coverage 3/4.

Changed-case reasoning: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4. Only the added recorded value contributes; the existing absent **field** remains skipped.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

▼ Support and alternative plan

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Lay out a row card beside the **accumulator**. Extension: count **process**ed records separately and report that count with the total.

▼ **Earlier wording, alternatives and picture transcripts**

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Preserve expected, observed and paper-trace records.

Supplied case materials and checked guidance

W2 running totals 2,6,7; reset inside loop wrongly leaves 1. Required missing used raises KeyError. Count policy includes 2 and 0, mean 1.0 over 2 observations/3 rows; after added 4, mean 2.0 over 3/4. All-empty mean None, observed sum 0 not unknown population total. Selected scores 2/6/missing give 8 2 3 4.0; three empty rows give 0 0 3 None. Preserve score coverage 2/3 and no fabricated missing scores.

Diagnose misconceptions

A successful run proves the conclusion: This loop assumes every record has the required integer **field**s. A missing used **field** causes KeyError. Do not silently treat a missing count as zero; that would invent unused capacity.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: records (7, 2), (3, 3), (8, 6) contribute 5, 0, 2 and total 7; reversed order also totals 7.

A convenient shortcut preserves meaning: Putting total = 0 inside the loop discards earlier contributions. With the original rows it finishes at 0, the second contribution.

Move the initialization before the loop and rerun both the original and reversed row order.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected bridge transfer task

> If a learner has seen this exact case, use a fresh parallel input and record exposure/support.

Selected independent task and checked key

Selected case: Trace the named **field**s and missing-value policy. Predict the **output**, state the observed total and mean denominator, and report coverage. What changes if all three rows are empty records?

```
rows = [{"score": 2}, {"score": 6}, {}]
values = []
for row in rows:
    value = row.get("score")
    if value is not None:
        values.append(value)
```

```
observed_total = sum(values)
if len(values) > 0:
    mean = observed_total / len(values)
else:
    mean = None
print(observed_total, len(values), len(rows), mean)
```

Output8 2 3 4.0: observed total8, two supplied scores out of three records, mean4 over denominator2.

With three empty records, **output**0 0 3 None: the empty observed sum is0 by the supplied summing procedure, coverage0/3, and no observed mean exists.

It does not claim the missing scores were zero or that the true total is known.

Judge trace and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Read a row

In the worked **dataset**, state the number of records and the spare spaces in the second record.

There are 2 records. The second has 0 spare spaces.

W2 Trace all updates

For spare contributions 2, 4 and 1, trace total starting at 0.

Running totals 2, 6, 7.

W3 Repair initialization

A loop resets total to 0 on each pass over contributions 2, 4, 1. What final total appears and how should it be fixed?

It ends at 1. Initialize total once before the loop; repaired total 7.

W5 Interpret an empty result

An empty **dataset** produces total 0. Can you conclude that every cupboard is full?

No. No records were **process**ed; the result provides no **evidence** about actual cupboards.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Contributions 5, 0, 6; total 11. Exit: No. No records were **process**ed; the result provides no **evidence** about actual cupboards.

W1 Read a row

W2 Trace all updates

W3 Repair initialization

W4 Process new records independently

Write a loop for records (capacity 9, used 2), (capacity 5, used 1), and (capacity 4, used 4). Predict the total. Reverse the order and test an empty list.

Contributions 7, 4, 0 give 11 in either order. Empty list gives 0. The loop must use both **field**s for every row.

W5 Interpret an empty result

Independent assessment and fresh retry

Independent W4 **evidence**: Contributions 7, 4, 0 give 11 in either order. Empty list gives 0. The loop must use both **field**s for every row. Record whether the learner needed no help, a hint or explanation.

Fresh retry: records (7, 2), (3, 3), (8, 6) contribute 5, 0, 2 and total 7; reversed order also totals 7.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Keep the portfolio, support and delayed check

> Track code execution separately from a correct paper trace.

Project evidence

Add a labelled reading structured data **evidence** sheet to Python readiness portfolio. Show how you can **process** a list of records consistently.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Support and selected assessment guidance

Lay out a row card beside the **accumulator**. Extension: count **process**ed records separately and report that count with the total.

Selected independent answer: Output8 2 3 4.0: observed total8, two supplied scores out of three records, mean4 over denominator2. With three empty records, output0 0 3 None: the empty observed sum is0 by the supplied summing procedure, coverage0/3, and no observed mean exists. It does not claim the missing scores were zero or that the true total is known.

Keep earlier W4 transfer as additional practice; do not substitute its numbers or key for this selected task. Record prior exposure and use a fresh parallel case after support.

Delayed return

Suggested return points in this level: python-bridge/12. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	

Record	Teacher evidence
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to the next topic

Next week: Testing and debugging functions. Reuse this week's explicit contracts and **evidence**.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the teaching cycle.

Planning and assessment reference

Teaching target: **Process** a list of records consistently

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: For spare contributions 2, 4 and 1, trace total starting at 0.

Reasoned practice answer: Running totals 2, 6, 7.

Check every learner before discussion: Use the worked case for Reading structured data. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: This loop assumes every record has the required integer **field**s. A missing used **field** causes KeyError. Do not silently treat a missing count as zero; that would invent unused capacity.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add {"count":4} as a fourth row. Require a **prediction** and a reason before comparing. Expected result: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

Reason: Only the added recorded value contributes; the existing absent **field** remains skipped.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep individual first attempts separate from supported rehearsal.

Feedback and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

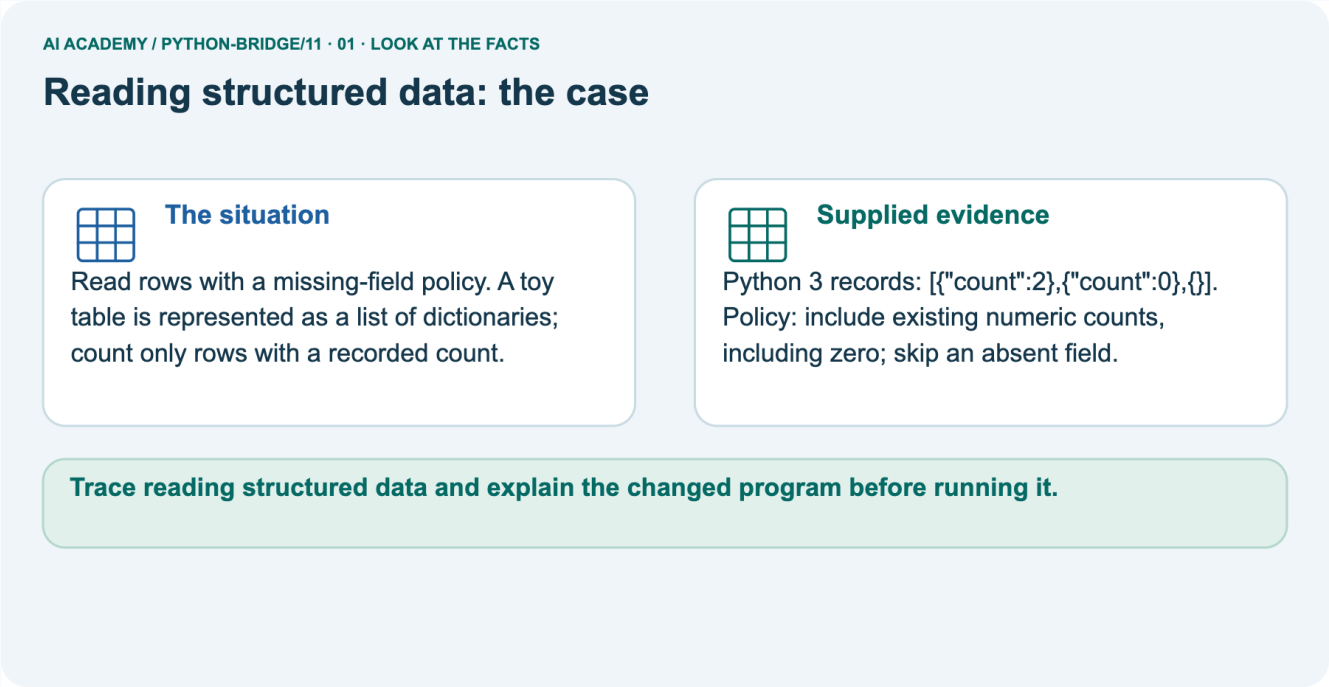
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Use the relevant picture for a specific reasoning move.

01. Reading structured data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

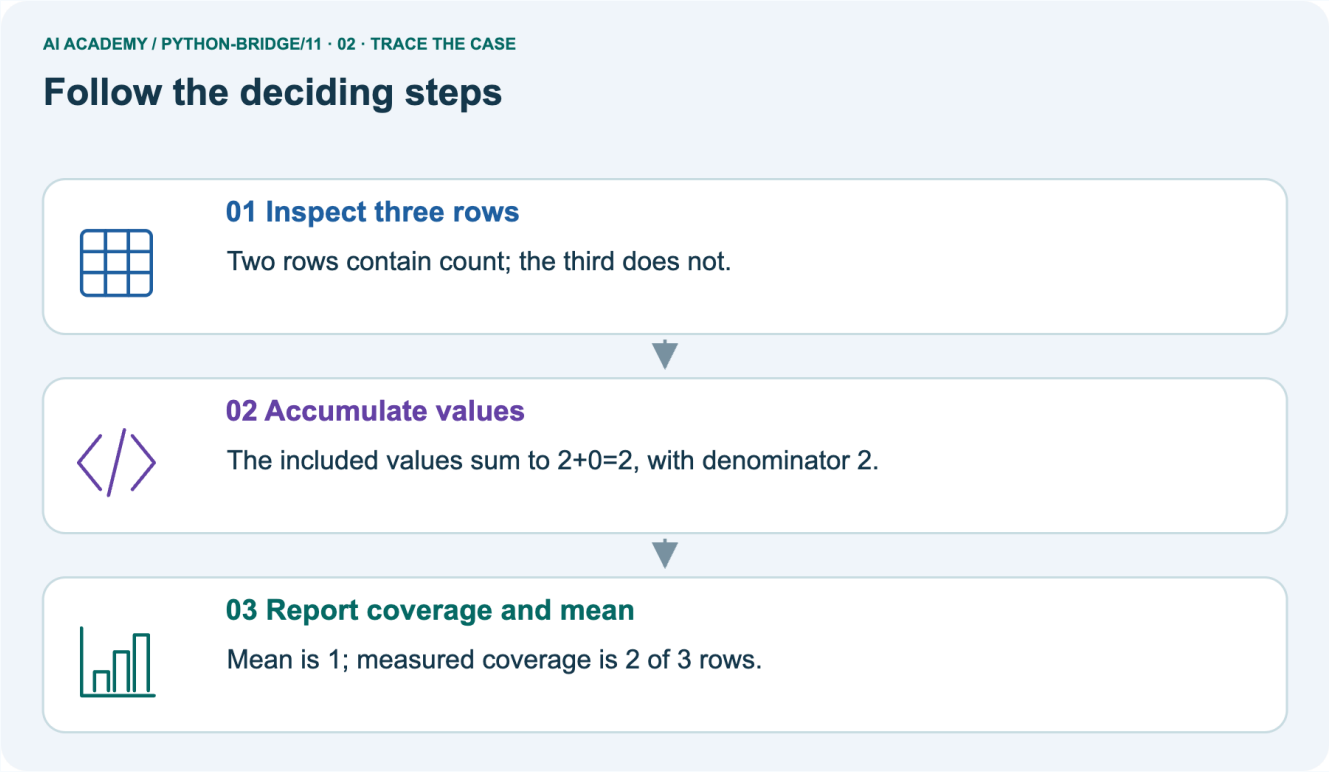
Read rows with a missing-**field** policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count. Supplied facts: Python 3 records: [{"count":2}, {"count":0},{}].

Policy: include existing numeric counts, including zero; skip an absent **field**.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



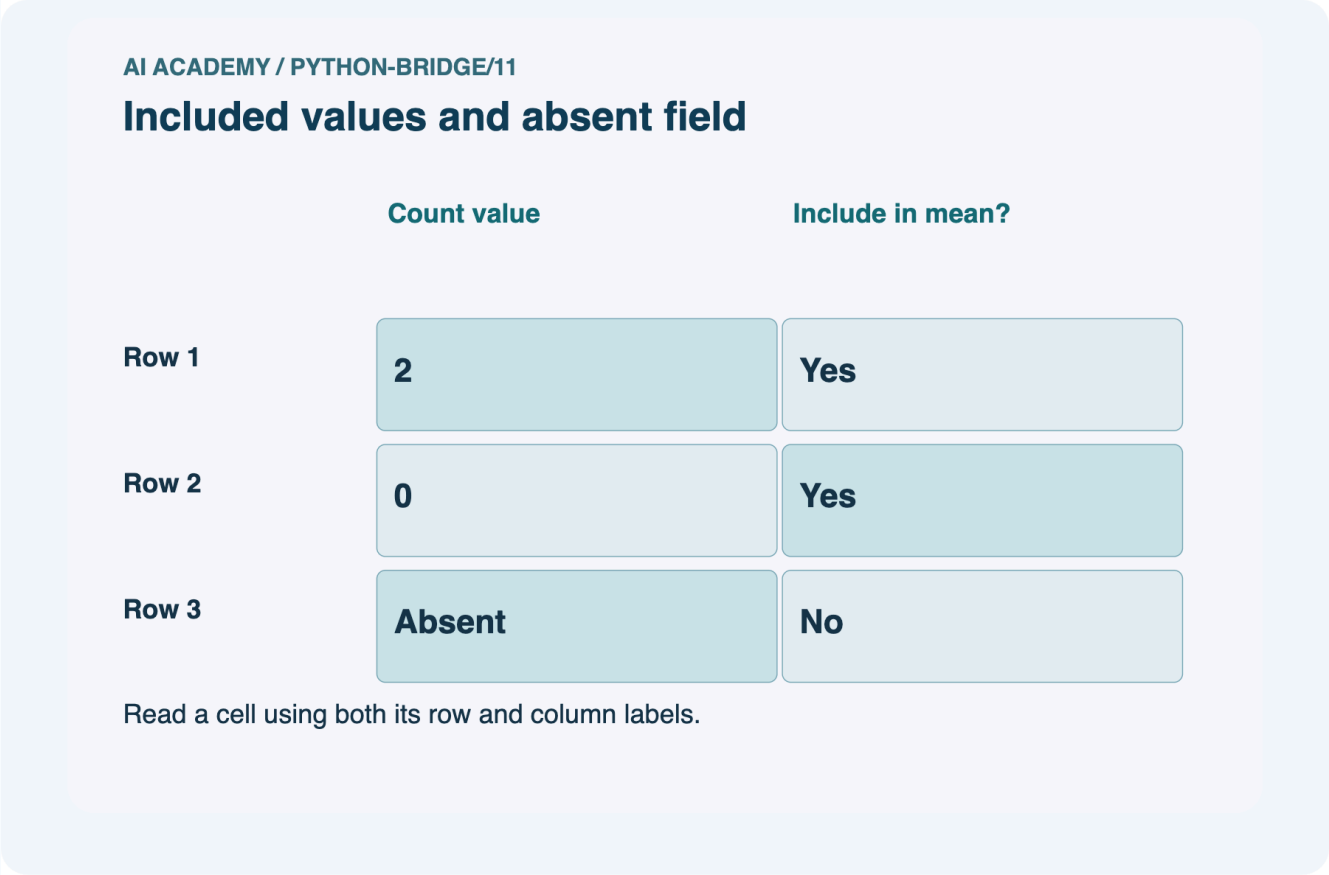
Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect three rows: Two rows contain count; the third does not. → Accumulate values: The included values sum to 2+0=2, with denominator 2. → Report coverage and mean: Mean is 1; measured coverage is 2 of 3 rows.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Included values and absent field



Follow the labelled picture with the matching explanation. Open the vector diagram

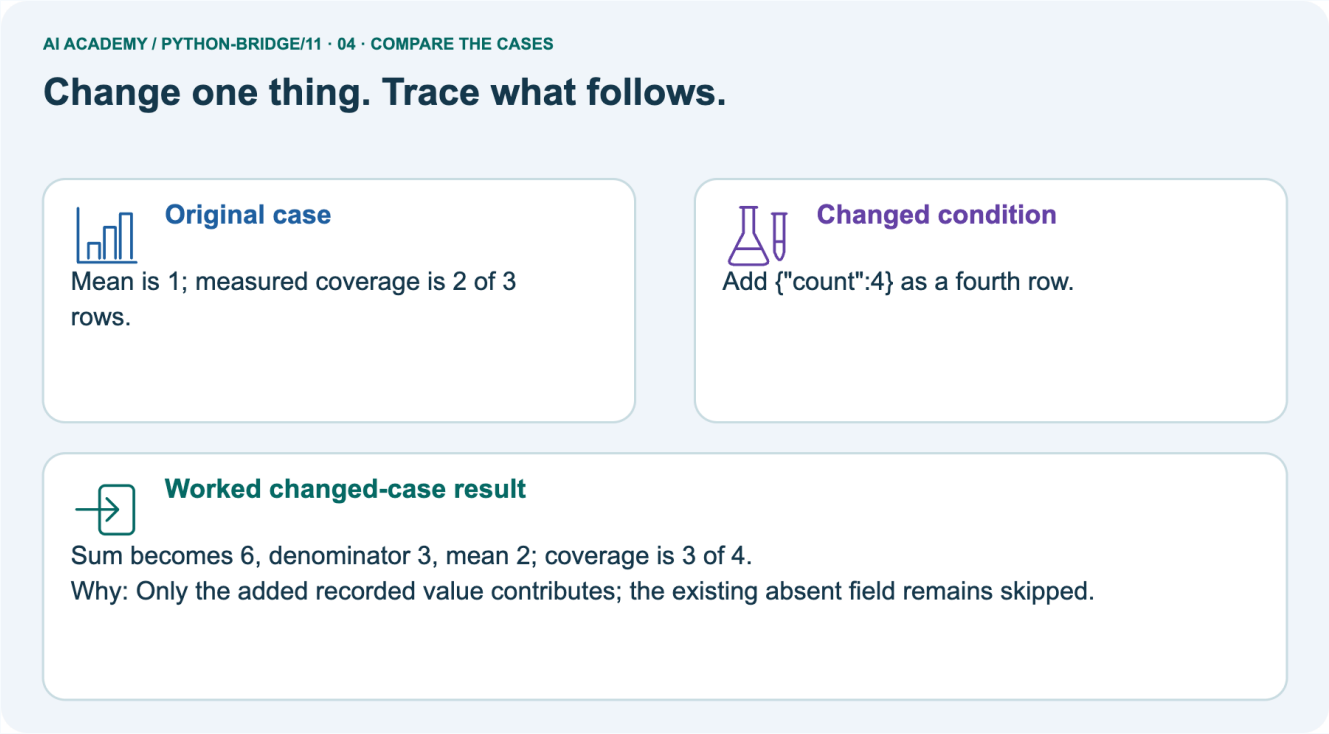
Included values and absent **field**. Row 1: Count value = 2, Include in mean? = Yes. Row 2: Count value = 0, Include in mean? = Yes. Row 3: Count value = Absent, Include in mean? = No.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Mean is 1; measured coverage is 2 of 3 rows. Changed condition: Add {"count":4} as a fourth row. Expected result: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

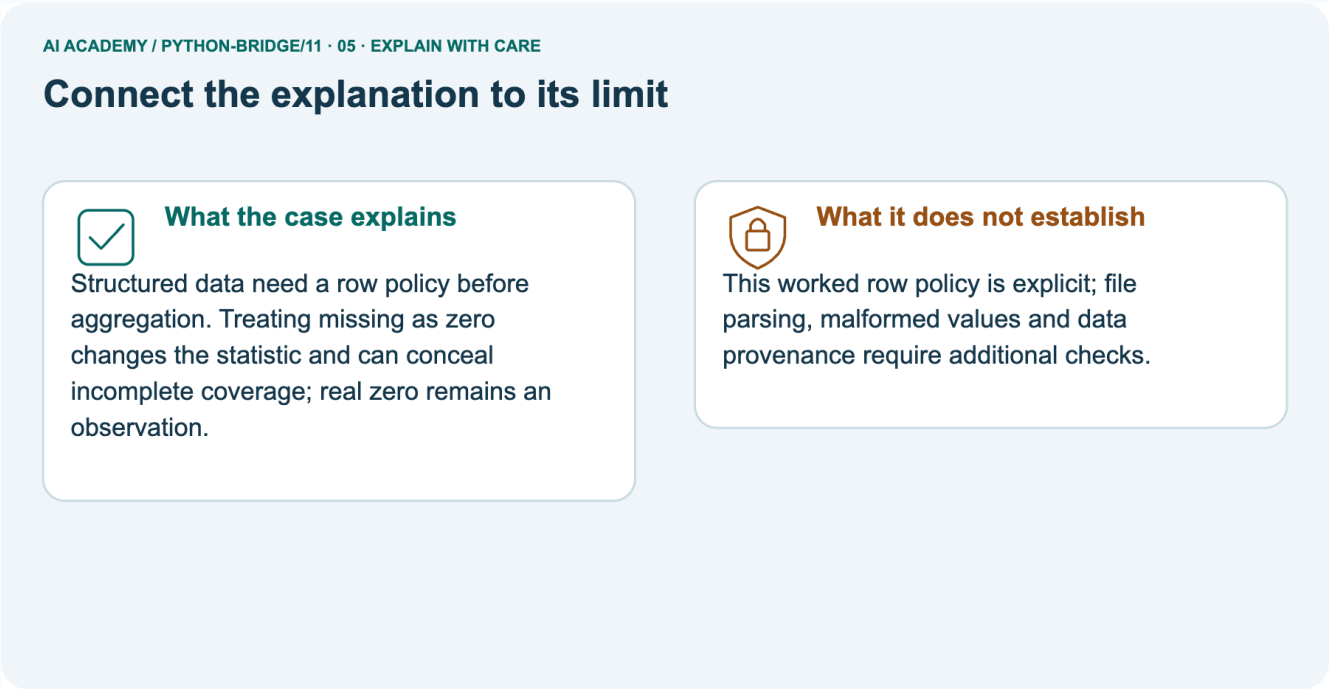
Reason: Only the added recorded value contributes; the existing absent **field** remains skipped.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Only the added recorded value contributes; the existing absent **field** remains skipped.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Structured data need a row policy before aggregation. Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an **observation**.

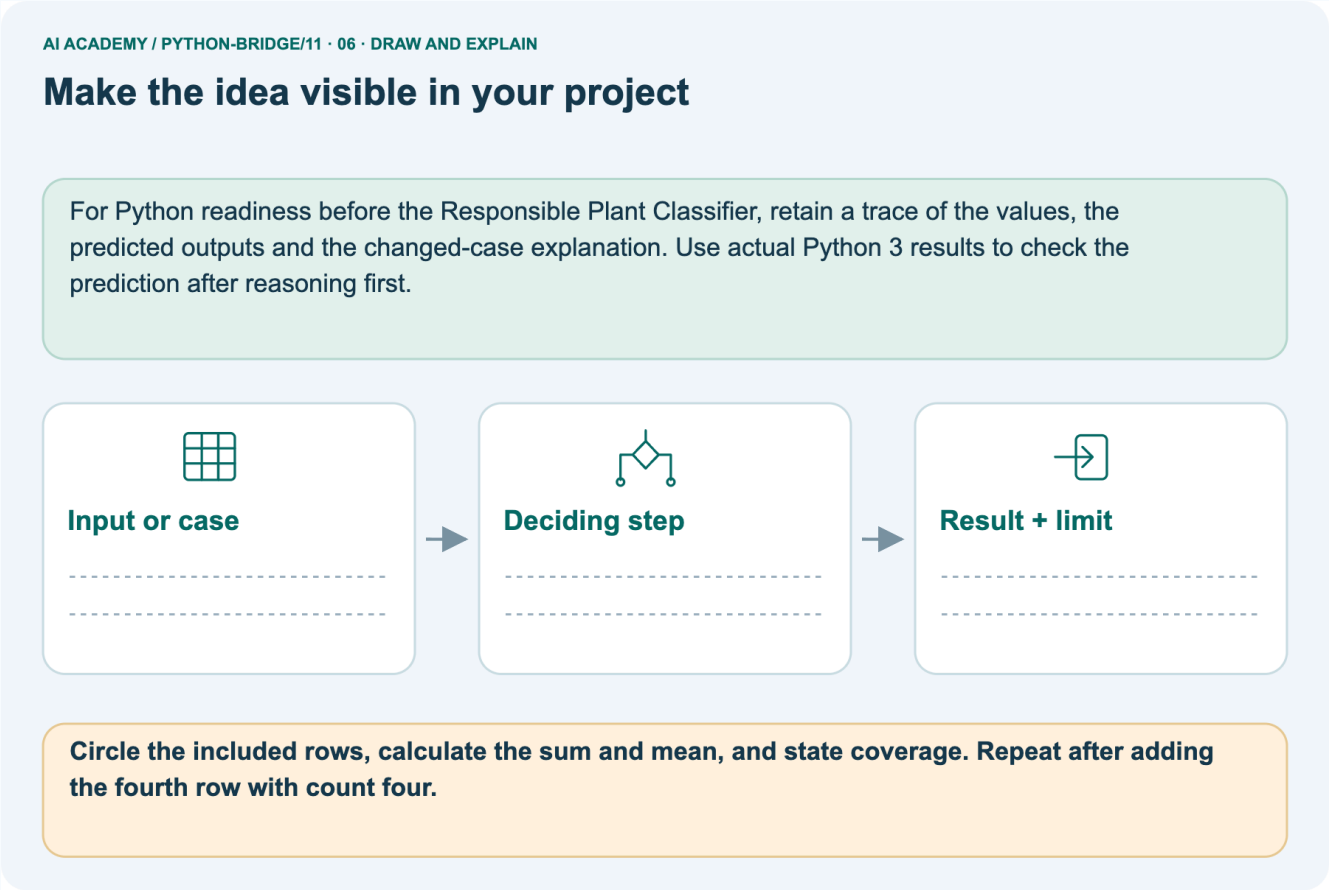
Limit: This worked row policy is explicit; file parsing, malformed values and data provenance require additional checks.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This worked row policy is explicit; file parsing, malformed values and data provenance require additional checks.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Initially values 2 and 0 give sum 2 and mean 1 across two **observation**s, coverage 2/3. Adding 4 gives sum 6, mean 2 across three **observation**s, coverage 3/4.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Reading structured data: the case

02. Follow the deciding steps

03. Included values and absent field

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.